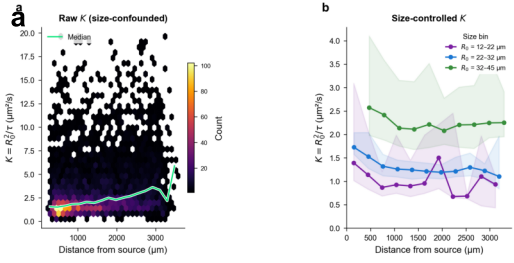
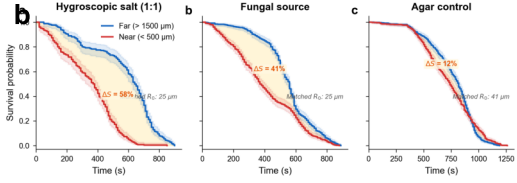
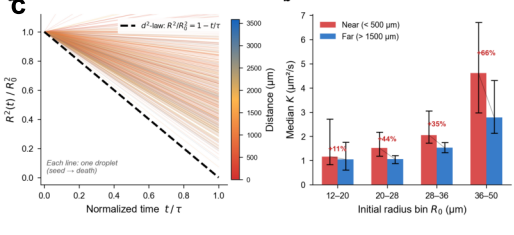
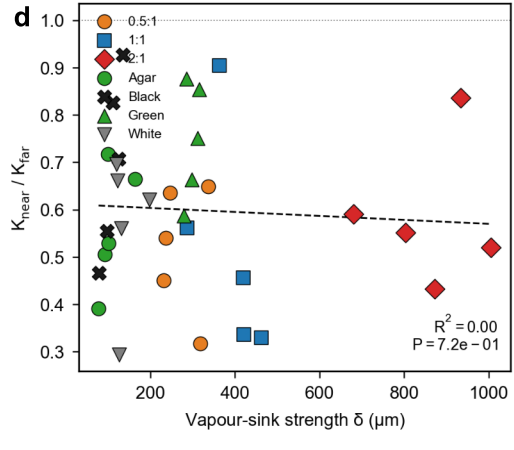

a
Raw K (size-confounded)
Median
K = R0²/τ (μm²/s)
Distance from source (μm)
Count
b
Size-controlled K
Size bin
R0 = 12–22 μm
R0 = 22–32 μm
R0 = 32–45 μm
K = R0²/τ (μm²/s)
Distance from source (μm)


b
Hygroscopic salt (1:1)
Far (> 1500 μm)
Near (< 500 μm)
ΔS = 58%
Matched R0: 25 μm
Survival probability
Time (s)
b
Fungal source
ΔS = 41%
Matched R0: 25 μm
Time (s)
c
Agar control
ΔS = 12%
Matched R0: 41 μm
Time (s)


a
c
d²-law: R²/R0² = 1 − tτ
Each line: one droplet (seed → death)
R²(t) / R0²
Normalized time t / τ
Distance (μm)
b
Near (< 500 μm)
Far (> 1500 μm)
+11%
+44%
+35%
+66%
Median K (μm²/s)
12–20
20–28
28–36
36–50
Initial radius bin R0 (μm)


d
0.5:1
1:1
2:1
Agar
Black
Green
White
Knear / Kfar
R² = 0.00
P = 7.2e − 01
Vapour-sink strength δ (μm)